

EE6347 - Devices for AI and Neuromorphic Computing

Tutorial 5

Task 1: Choosing V_{write} for programming a passive RRAM

- You can use the same models used in the mid-semester exam. You can use **1V** for the pulses and use a pulse-width of **11 ms** (which would roughly give a 1.3 uS conductance in the mid-sem)
- Find how much conductance a $V/2$ pulse potentiates the RRAM cell to
- Thus find the ratio of conductance between full selected and half selected cells

Task 2: Sneak Currents when using $V/2$ Scheme

- Build a 3x3 Passive Crossbar with lprobes in series within each cell.
- Set voltage sources on all the lines and write the (0,0) cell.
- Measure the current through the current through (1,0) cell.
- Measure the current through the (2,2) cell.

Task 3: Sneak Currents when using $V/3$ Scheme

- Now use the $V/3$ scheme. Measure current through (1,0) cell. Compare this with the $V/2$ scheme
- Now measure the (2,2) cell. Compare this with the $V/2$ scheme

Task 4: Sneak Currents when the undesired lines are floated

- Now remove all the voltage sources on the half-selected and unselected cells and float them.
- Write the (0,0) cell, measure current through (1,1) cell.
- Now in the same testbench, write (1,1) cell and measure the current through (1,0) cell.
- Why is this different from what you measured in part b? Explain why we use a $V/2$ or a $V/3$ scheme as opposed to just floating the unused lines based on this observation